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Power Flow Surrogate for Power Systems with High Renewable Penetration via a Physics-Informed Graph Attention Network

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Abstract

The increasing integration of renewable generation introduces highly stochastic operating conditions, substantially enlarging the operating space and posing severe computational challenges for traditional iterative power flow solvers. To address this, we propose a Physics-Informed Graph Attention Network (PI-GAT) for fast and physically consistent power flow assessment in power systems with high renewable penetration. PI-GAT represents buses and branches as graph-structured inputs and employs edge-aware multi-head attention to adaptively capture electrical interactions between connected nodes. By embedding AC power flow equations as residuals in the training loss, PI-GAT promotes physical consistency, improving nodal power balance consistency even under high renewable variability and N−1 contingency scenarios. Experimental results on IEEE 30-bus and 118-bus systems demonstrate that PI-GAT reduces active and reactive power mismatches by up to approximately 62% across the two benchmark systems relative to the edge-aware GAT baseline. This improvement in physical consistency is accompanied by a modest increase in point-wise voltage and phase-angle errors. Moreover, PI-GAT achieves substantial inference-time speedups over conventional numerical solvers, especially under batched multi-scenario inference. These findings indicate that PI-GAT provides a reliable and efficient surrogate model for real-time security assessment and contingency screening in power systems with high penetration of renewable generation.

Keywords: graph attention network; physics-informed neural network; power flow; renewable energy

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1. Introduction

The increasing integration of renewable energy sources (RES), especially wind and photovoltaic (PV) generation, has fundamentally altered the operational landscape of modern power systems [1,2]. Unlike conventional synchronous generators, whose outputs can be scheduled within relatively well-defined dispatch ranges, wind and PV generation are strongly affected by weather-dependent variability and forecasting uncertainty [3–6]. This stochastic nature substantially expands the range of feasible operating conditions within the system, posing significant challenges to voltage regulation, frequency stability, and power balance across the network. As renewable penetration continues to grow in response to decarbonization targets, grid operators must evaluate an enormous number of operating scenarios in real-time to maintain security and stability, particularly under frequent fluctuations in power injections [7–9].

Conventional power flow solvers, such as Newton–Raphson [10] and fast-decoupled methods [11], remain the standard tools for computing steady-state voltage and power distributions in transmission networks, offering well-established convergence guarantees and high numerical accuracy [12–14]. However, these methods are inherently iterative. At each step, the system of nonlinear power balance equations is linearized around the current voltage estimate, and the process is repeated until the power mismatch falls below a prescribed tolerance. While this approach is reliable for individual scenario evaluation, its computational cost scales poorly with both system size and the number of scenarios to be assessed, making it increasingly burdensome in large-scale operational environments [15–17]. In online security assessment, operators may need to screen thousands of contingency scenarios within seconds to make timely operational decisions [18–20]. As renewable penetration rises and operating conditions become more diverse, the number of probabilistic scenarios to evaluate grows dramatically, making it increasingly difficult to meet the real-time dispatch and control requirements with conventional solvers alone. These limitations have motivated growing interest in the development of data-driven surrogate models that approximate power flow solutions with significantly reduced computational effort, enabling faster and more scalable scenario evaluation without sacrificing much accuracy.

Data-driven machine learning approaches have emerged as promising alternatives for accelerating power flow computation, offering the prospect of a near-instantaneous solution once a model is trained offline. Early work demonstrated that multilayer perceptrons (MLPs) could map bus power injections to voltage solutions with significant speedups over iterative solvers, though generalization across varying topologies remained limited [21,22]. Among more recent approaches, graph neural networks (GNNs) are particularly well-suited for power systems because they naturally capture the graph-structured topology of buses and branches, allowing information to propagate along physical connections in a manner that directly mirrors the locality of power flow equations [23,24]. Graph convolutional networks (GCNs) have demonstrated competitive accuracy in approximating power flow solutions and evaluating system stability across standard benchmark systems [25–28]. Building on this, graph attention networks (GATs) introduce learnable attention coefficients that adaptively weight the contributions of neighboring nodes, enabling more expressive representations of heterogeneous network interactions and improving prediction fidelity under varying operating conditions [29–31].

However, most purely data-driven GNN or GAT surrogates are trained mainly by minimizing prediction errors with respect to numerical power flow labels. As a result, small voltage-magnitude errors do not necessarily imply that the predicted voltage phasors satisfy the AC power flow equations. When the predicted voltages are substituted back into the nodal power-balance equations, non-negligible active and reactive power mismatches may still occur [32]. This issue is particularly important for real-time security assessment and contingency screening, where physically inconsistent surrogate predictions may lead to misleading operational decisions even if their point-wise voltage errors appear acceptable.

Physics-informed neural networks (PINNs) provide a natural way to improve the physical consistency of neural-network predictions by embedding governing equations or residual constraints into the training loss [33]. In power and energy systems, physics-informed learning has been explored for state estimation [34], transient stability assessment [35], load margin assessment [36], contingency screening [37], fault location [38], dynamic parameter identification [39], and thermal anomaly prediction [40]. Power-flow-related studies have also incorporated Kirchhoff-law-based constraints, DC/AC optimal power flow constraints, or power flow residuals into neural-network training to reduce infeasible predictions [41–43]. Hu et al. proposed a physics-guided deep neural network for power flow analysis, in which Kirchhoff-law-related physical information and system topology

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were introduced to improve the generalization of data-driven power flow solvers [41]. Nellikkath and Chatzivasileiadis further applied physics-informed neural networks to DC and AC optimal power flow, demonstrating that embedding OPF and power flow constraints into neural-network training can reduce constraint violations and improve the feasibility of predicted solutions [42,43]. These studies show that physics-informed learning can improve the physical consistency of neural power-system surrogates.

Nevertheless, many PINN-based power flow or OPF formulations introduce physical knowledge mainly through the training loss, while the neural-network architecture itself does not always explicitly exploit branch-level graph structure or topology-dependent electrical interactions. In particular, branch admittances, transformer tap ratios, and branch operating status are not always directly used in the message-passing or attention mechanism. This limitation may weaken the ability of the surrogate model to distinguish different electrical couplings and topology-dependent operating conditions.

More recently, physics-informed graph-based models have been investigated for power flow calculation and contingency-related tasks. For example, Jeddi and Shafieezadeh proposed a physics-informed graph attention-based approach for power flow analysis [44], while Zhang et al. developed a physics-informed line graph neural network that propagates information between buses and transmission lines for power flow calculation [45]. These studies combine graph representation with physical regularization and provide important foundations for physics-informed graph learning in power systems. However, existing studies usually emphasize only part of the requirements for fast and reliable power flow assessment. Some methods focus on graph-based representation but do not enforce AC power flow residuals. Some incorporate physical constraints but do not use branch electrical parameters directly in the attention mechanism. Others consider topology changes but require graph reconstruction or do not explicitly encode branch outage status in a fixed graph. Therefore, the joint integration of branch-level electrical feature modeling, bus-type-aware AC residual regularization, and fixed-graph N−1 contingency encoding remains insufficiently explored.

To address these limitations, this paper proposes a physics-informed graph attention network, termed PI-GAT, for fast and physically consistent AC power flow prediction. The proposed model is designed around three key considerations. First, branch electrical parameters are incorporated into the attention mechanism so that message passing is not only topology-aware but also edge-aware. Second, AC power flow residuals are introduced into the training objective to penalize physically inconsistent voltage predictions. Third, branch-status variables are embedded into edge features, allowing normal and N−1 line-outage scenarios to be represented on a fixed graph without repeated graph reconstruction.

To better position PI-GAT with respect to existing learning-based power flow surrogate models, Table 1 summarizes the main differences among representative methods. As shown in the table, existing studies usually focus on only part of the above requirements, such as graph-based representation, attention-based aggregation, or physics-informed regularization. In contrast, PI-GAT jointly integrates edge-aware attention, AC residual regularization, and fixed-graph contingency encoding in a unified surrogate model. The main contributions are summarized as follows:

- An edge-aware multi-head graph attention mechanism is designed to incorporate branch electrical parameters directly into the attention computation. Multiple attention heads independently learn to capture different aspects of the electrical interactions between connected buses, while each head jointly accounts for the states of both endpoint buses and the physical parameters of the connecting branch, enabling the model to modulate neighbor influence according to impedance characteristics and operational status. Under N−1 contingency conditions, this design allows the attention mechanism to learn to suppress the contribution of faulted lines through the branch status feature, without requiring dynamic graph recomputation at inference time.

- A physics-informed graph attention network (PI-GAT) is proposed that embeds AC power flow residuals as explicit penalty terms in the training loss. By substituting predicted voltages back into the governing power flow equations, the physics residual guides the predicted solutions toward lower nodal power-balance residuals without requiring additional labeled data. A warm-up-ramp scheduling strategy is further introduced to stabilize training across systems of different scales, preventing destabilizing gradient signals in early training when predicted voltages are far from the true solution.
- Comprehensive experiments on the IEEE 30-bus and IEEE 118-bus systems demonstrate the effectiveness and efficiency of PI-GAT. PI-GAT reduces active and reactive power mismatch by up to approximately 62% relative to the edge-aware GAT baseline while maintaining voltage errors within the evaluated tolerance range, with the physical consistency advantage consistently preserved across both normal operating and N−1 contingency scenarios, as verified through scenario-separated evaluation on both benchmark systems. Furthermore, PI-GAT achieves clear single-scenario and batched inference-time speedups over conventional numerical solvers, confirming its potential for repeated multi-scenario contingency screening and dispatch-support applications.

Table 1. Comparison of PI-GAT with representative learning-based power flow surrogate studies.

Studies	**Main Focus**	**Edge-Aware Modeling**	**AC Residual**	**N−1/Topology Handling**
MLP-based PF surrogates [21,22]	Injection-to-voltage mapping	No graph structure	No	Not considered
GCN-based PF surrogates [23,25,26]	Topology-aware voltage prediction	Limited	No	Topology-dependent graph
GAT-based PF surrogates [30,31]	Attention-based aggregation	Partial	No	Topology-dependent graph
PINN-based PF and OPF surrogates [41–43]	PF residual regularization	Limited	Yes	Task-dependent
Physics-informed graph models [32,44,45]	Graph learning with physics	Method-dependent	Yes/partial	Method-dependent
Proposed PI-GAT	Edge-/physics-/contingency-aware PF surrogate	**Branch parameters in attention**	**Bus-type-aware AC residual**	**Branch status on fixed graph**

Note: The bolded row corresponds to the PI-GAT method proposed in this work, while all other rows summarize existing research from previous literature. Bold formatting is adopted to clearly distinguish our novel approach from prior power flow surrogate models listed in the table.

The rest of the paper is organized as follows. Section 2 formulates the power system graph model and the AC power flow equations. Section 3 details the proposed PI-GAT framework, including feature construction, the edge-aware graph attention architecture, and the physics-informed training strategy. Section 4 describes the experimental setup and discusses the results on both benchmark systems. Section 5 concludes the paper.

2. Problem Formulation and Representation

This section establishes the mathematical and structural foundation of the proposed framework. The AC power flow equations are first presented to formalize the physical constraints governing steady-state operation of the power network. The power system topology is then abstracted as a graph, where buses and branches are represented as nodes and edges, and admittance parameters are incorporated as edge attributes, providing the structural basis for the graph attention network architecture.

2.1. AC Power Flow Model

The steady-state operation of the network is governed by the AC power flow equations, which express the active and reactive power balance at each bus i as

$$\begin{aligned} P_i &= V_i \Sigma_{j \in \overline{\mathcal{N}}_i} V_j (G_{ij} cos\theta_{ij} + B_{ij} sin\theta_{ij}) \\ Q_i &= V_i \Sigma_{j \in \overline{\mathcal{N}}_i} V_j (G_{ij} sin\theta_{ij} - B_{ij} cos\theta_{ij}) \end{aligned} \tag{1}$$

Here, V_i and θ_i denote the voltage magnitude and phase angle at bus i, respectively, and $\theta_{ij} = \theta_i - \theta_j$. Let $Y_{ij} = G_{ij} + \mathrm{j}B_{ij}$ denote the (i, j)-th element of the bus-admittance matrix $\mathbf{Y}$, where G_{ij} and B_{ij} are its real and imaginary parts, respectively. The set $\overline{\mathcal{N}}_i = \mathcal{N}_i \cup \{i\}$ contains bus i and all buses directly connected to it, so that the diagonal self-admittance term is included in the power flow summation. The slack-bus voltage magnitude is fixed at its specified setpoint, while its phase angle serves as the angular reference.

Equation (1) is traditionally solved via iterative numerical methods, most commonly the Newton–Raphson algorithm, which linearizes the power mismatch around the current voltage estimate at each step until convergence is achieved. While the Newton–Raphson algorithm offers reliable quadratic convergence under well-conditioned operating conditions, it requires the repeated construction and factorization of a Jacobian matrix for each scenario, resulting in computational costs that scale poorly with network size. In large-scale systems subject to high renewable variability, where thousands of diverse scenarios must be evaluated within operational time windows, these limitations become increasingly prohibitive. This motivates the development of a learning-based surrogate capable of approximating power flow solutions in a single non-iterative forward pass, while maintaining physical consistency with the governing equations.

2.2. Graph Representation of the Power Network

The power grid topology is naturally modeled as an undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V} = \{1, 2, \ldots, N\}$ denotes the set of N buses and $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ denotes the set of transmission branches and transformers. Although the physical network is topologically undirected, each branch is represented by two directed message-passing edges, $j \to i$ and $i \to j$, in the implementation. Each node $i \in \mathcal{V}$ is associated with a feature vector $\mathbf{x}_i \in \mathbb{R}^{d_v}$ encoding its operational state, and each edge $(i, j) \in \mathcal{E}$ is associated with a feature vector $\mathbf{e}_{ij} \in \mathbb{R}^{d_e}$ encoding the physical parameters of the connecting branch. The specific construction of these feature vectors is described in Section 3.

The bus-admittance matrix $\mathbf{Y} \in \mathbb{C}^{N \times N}$ governs the electrical coupling among buses. For a standard transmission line (i, j), let $y_{ij} = g_{ij} + \mathrm{j}b_{ij} = 1/(r_{ij} + \mathrm{j}x_{ij})$ denote its series admittance. In the absence of an off-nominal transformer tap, the corresponding off-diagonal bus-admittance entries satisfy ($Y_{ij} = Y_{ji} = -y_{ij}$). For transformer branches, these entries are modified according to the complex tap ratio. Let $y_s = 1/(r + \mathrm{j}x)$ denote the branch series admittance, b_c denote the total line-charging susceptance, and $t = \tau e^{\mathrm{j}\phi}$ denote the complex off-nominal transformer tap ratio located at the i-bus side. The corresponding branch contributions to the bus-admittance matrix are

$$\begin{aligned} Y_{ii} &\leftarrow Y_{ii} + \frac{y_s + \mathrm{j}b_c/2}{|t|^2}, \\ Y_{ij} &\leftarrow Y_{ij} - \frac{y_s}{t^*}, \\ Y_{ji} &\leftarrow Y_{ji} - \frac{y_s}{t}, \\ Y_{jj} &\leftarrow Y_{jj} + y_s + \mathrm{j}b_c/2. \end{aligned} \tag{2}$$

For a branch without an off-nominal transformer, $\tau = 1$ and $\phi = 0$.

The diagonal entry $Y_{ii} = G_{ii} + \mathrm{j}B_{ii}$ aggregates the contributions of all incident branches, line charging, transformer parameters, and bus shunt elements. The magnitude $|Y_{ij}|$ quantifies the strength of the electrical interaction between the two endpoints, naturally playing the role of an edge weight in the graph.

The structural correspondence between the power flow equations and graph-based computation motivates a neighborhood-aggregation representation of the network interactions

$$\mathbf{s}_i = \phi\left(\mathbf{x}_i, \bigoplus_{j \in \mathcal{N}_i} \psi(\mathbf{x}_i, \mathbf{x}_j, \mathbf{e}_{ij})\right) \tag{3}$$

where ψ computes a message from neighbor j to bus i using both node states and edge parameters, $\bigoplus$ denotes a permutation-invariant aggregation operator, and ϕ updates the state of bus i based on the aggregated neighborhood information. This formulation establishes a direct structural correspondence with Equation (1). The power injection at bus i is determined by aggregating contributions from its electrical neighbors, weighted by the branch admittances G_{ij} and B_{ij}. The power flow problem thus admits a natural graph computational interpretation, motivating the use of graph neural networks as a principled architectural choice for surrogate modeling.

The AC power flow equations show that the electrical interaction between connected buses depends on the corresponding entries of the bus-admittance matrix. These entries are determined by the physical parameters of the connecting branches, including their series impedances, shunt charging, and transformer tap ratios. This observation motivates the use of edge-aware graph attention, in which branch-level electrical parameters are incorporated directly into the message-passing mechanism so that the model can assign different weights to neighboring buses according to the physical characteristics of each connecting branch.

3. Physics-Informed Graph Attention Network

Building on the graph formulation established in Section 2, this section presents the architecture and training objective of the proposed PI-GAT framework. The model takes the graph-structured representation of the power network as input and produces bus voltage magnitudes and phase angles as output. The architecture leverages multi-head graph attention to propagate electrical state information along the network topology, while a physics-informed loss function embeds AC power flow residuals directly into the training objective to improve physical consistency of the predicted solutions.

3.1. Feature Construction

The input to PI-GAT consists of the node feature matrix $\mathbf{X} \in \mathbb{R}^{N \times 8}$ and the edge feature matrix $\mathbf{E} \in \mathbb{R}^{|\mathcal{E}| \times 7}$, constructed from the graph representation defined in Section 2.2. The feature design aims to provide the model with the complete operational context necessary to infer the voltage solution at each bus, while explicitly encoding the structural distinctions between different generation types and bus roles.

Each node feature vector $\mathbf{x}_i \in \mathbb{R}^8$ encodes the operational state of bus i as follows:

- P_i^{load}, Q_i^{load}: active and reactive load demand;
- P_i^{ren}: active power output of renewable generation;
- P_i^{gen}: active power output of conventional dispatchable generation;
- V_i^{set}: voltage magnitude setpoint;
- $is_i^{PQ}, is_i^{PV}, is_i^{ref}$: one-hot binary indicators encoding bus type.

Separating renewable injection P_i^{ren} from conventional generation P_i^{gen} as distinct input channels reflects their fundamentally different operational characteristics. Conventional units operate within narrow dispatch bounds defined by generator capacity constraints, whereas renewable sources exhibit wide and stochastic output variability that can range from zero to rated capacity within short time intervals. This separation is particularly suitable for power systems with high renewable penetration, where heterogeneous distributed energy resources ranging from dispatchable units to stochastic renewables coexist and exhibit fundamentally distinct operational characteristics. The one-hot bus-type encoding explicitly informs the model of the role each bus plays in the power flow problem, distinguishing between PQ, PV, and slack buses, and directly mirrors the variable classification used in conventional power flow solvers, where different quantities are specified and inferred at each bus type.

Each edge feature vector $\mathbf{e}_{ij} \in \mathbb{R}^7$ encodes the physical parameters of branch (i, j):

- r_{ij}, x_{ij}: series resistance and reactance;
- sh_{ij}: shunt charging susceptance;
- tap_{ij}: transformer tap ratio, set to 1.0 for non-transformer branches;
- $status_{ij}$: branch in-service indicator;
- g_{ij}, b_{ij}: series conductance and susceptance derived from r_{ij} and x_{ij}.

The inclusion of both impedance parameters (r_{ij}, x_{ij}) and their derived admittance counterparts (g_{ij}, b_{ij}) provides the model with complementary representations of branch electrical characteristics, reducing the need to learn nonlinear transformations between these quantities. Here, the lowercase quantities g_{ij} and b_{ij} denote the real and imaginary parts of the branch series admittance y_{ij}, whereas the uppercase quantities G_{ij} and B_{ij} denote the real and imaginary parts of the bus-admittance-matrix entry Y_{ij} used in the AC power flow equations. The transformer tap ratio tap_{ij} allows the model to distinguish between transmission lines and transformer branches without requiring separate architectural components.

Under N−1 contingency conditions, the faulted branch is retained in the graph with $status_{ij} = 0$ rather than removed from $\mathcal{E}$. This design choice preserves the fixed graph structure required for efficient batched inference, while enabling the model to learn the effect of line outages implicitly from the training data. Encoding the outage as a feature rather than a structural modification avoids the need for dynamic graph recomputation at inference time, which would otherwise introduce significant overhead in real-time contingency screening applications. All node and edge features are standardized to zero mean and unit variance prior to training to ensure numerical stability and comparability across features of different physical units. The output of PI-GAT is a matrix $\hat{\mathbf{Z}} \in \mathbb{R}^{N \times 2}$, where each row $[\hat{V}_i, \ \hat{\theta}_i]$ contains the predicted voltage magnitude and phase angle at bus i. Voltage magnitudes and phase angles are not used as online input features; they serve only as supervised labels generated offline by the numerical power flow solver.

3.2. Graph Attention Network Architecture

The overall architecture of PI-GAT for power flow surrogate is illustrated in Figure 1, comprising a feature encoder, a stack of edge-aware graph attention layers, and an output decoder connected in a forward inference pipeline. During training, the predicted voltage solutions are substituted back into the AC power flow equations to compute the physics residual, which is incorporated as an additional loss signal alongside the data fidelity term to update the network weights.

The input encoder projects the raw node feature vector $x_i \in \mathbb{R}^8$ into a hidden representation $\mathbf{h}_i^0 \in \mathbb{R}^{d_h}$ through two successive linear layers, each followed by LayerNorm and SiLU activations. This two-layer design allows the model to learn an expressive initial representation of each bus before message passing begins, while the normalization ensures that features of different physical scales contribute comparably to the subsequent attention computation.

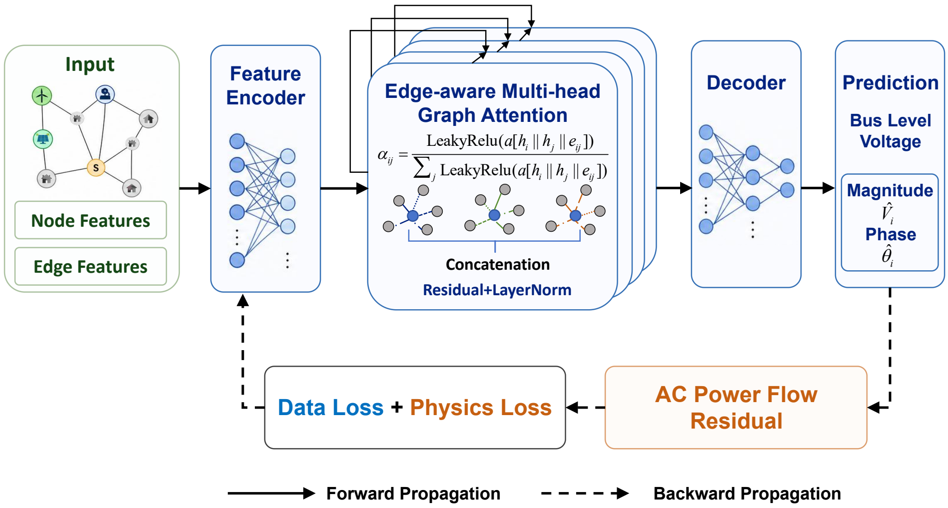


Figure 1. Overall architecture of the proposed PI-GAT framework.

Each graph attention layer computes edge-aware multi-head attention coefficients that jointly account for the states of both endpoint buses and the physical parameters of the connecting branch. For the k-th attention head at layer l, the attention coefficient from bus j to bus i is computed as

$$\alpha_{ij}^{(l,k)} = \mathrm{softmax}_j\left(\mathrm{LeakyReLU}\left(\mathbf{a}^{(l,k)} \cdot \left[\mathbf{W}^{(l,k)}\mathbf{h}_i^l \,\|\, \mathbf{W}^{(l,k)}\mathbf{h}_j^l \,\|\, \mathbf{W}_e^{(l,k)}\mathbf{e}_{ij}\right]\right)\right) \tag{4}$$

Here, $\mathbf{W}^{(l,k)} \in \mathbb{R}^{(d_h/K)\times d_h}$ and $W_e^{(l,k)} \in \mathbb{R}^{(d_h/K)\times 7}$ are learnable projection matrices for node and edge features respectively, $\mathbf{a}^{(l,k)} \in \mathbb{R}^{3d_h/K}$ is a learnable attention vector, K denotes the number of attention heads, and $\|$ denotes vector concatenation. The incorporation of the branch feature vector $\mathbf{e}_{ij}$ directly into the attention computation is a key design. It allows the model to adjust the influence of each neighbor according to the electrical characteristics of the connecting branch, such as impedance and operational status, rather than relying solely on node-level information. This is particularly important under N$-$1 contingency conditions, where $status_{ij} = 0$ signals the removal of a branch and the attention mechanism can learn to suppress the corresponding message.

The updated node representation at layer $l+1$ aggregates neighborhood information weighted by the computed attention coefficients, with outputs from all K heads concatenated to form the full hidden representation

$$\mathbf{h}_i^{l+1} = \Big\|_{k=1}^{K} \sum_{j\in\mathcal{N}_i} \alpha_{ij}^{(l,k)}\mathbf{W}^{(l,k)}\mathbf{h}_j^l \tag{5}$$

A residual connection is applied after ELU activation and dropout, followed by LayerNorm, to stabilize gradient flow and mitigate the over-smoothing phenomenon that causes node representations to become indistinguishable in deep graph networks

$$\mathbf{h}_i^{l+1} \leftarrow \mathrm{LayerNorm}\left(\mathrm{Dropout}\left(\mathrm{ELU}\left(\mathbf{h}_i^{l+1}\right)\right) + \mathbf{h}_i^l\right) \tag{6}$$

The output decoder maps the final node representations $\mathbf{h}_i^L$ to voltage predictions through a three-layer MLP with SiLU activations, producing a two-dimensional output for each bus. The predicted voltage magnitude $\hat{V}_i$ is obtained by adding 1.0 p.u. to the first output channel, encoding the prior that bus voltages in normal and mildly perturbed operation remain close to their nominal value. The predicted phase angle $\hat{\theta}_i$ is taken directly from the second output channel without additional transformation, as phase angles can take both positive and negative values relative to the slack bus reference and require no such initialization prior.

3.3. Physics-Informed Loss Function

A purely data-driven model minimizes the discrepancy between predicted and ground-truth voltage solutions, but provides no guarantee that predictions satisfy the underlying physical equations. Even when the mean prediction error is small, the substituted voltages may produce substantial power mismatches when evaluated against the AC power flow equations, rendering the predictions physically inadmissible for use in security assessment. To address this, the training objective of PI-GAT combines a supervised regression loss with a physics residual penalty

$$\mathcal{L} = \mathcal{L}_{MSE} + \lambda \mathcal{L}_{phys} \tag{7}$$

where $\lambda > 0$ is a scalar hyperparameter that controls the relative weighting between data fidelity and physical consistency. In practice, λ is tuned on the validation set and may differ across systems of different scales.

The data fidelity term $\mathcal{L}_{MSE}$ is the mean squared error between predicted and ground-truth voltage solutions across all N buses

$$\mathcal{L}_{MSE} = \frac{1}{N} \sum_{i=1}^{N} \Big((\hat{V}_i - V_i)^2 + (\hat{\theta}_i - \theta_i)^2 \Big) \tag{8}$$

Voltage phase angles are represented in radians during model training and physics-residual evaluation, and are converted to degrees only when the evaluation metrics are reported. This term supervises the model using labeled power flow solutions generated by conventional numerical solvers, ensuring that the predicted voltages remain close to the ground-truth operating state.

The physics residual term $\mathcal{L}_{phys}$ penalizes deviations from nodal power balance by substituting the predicted voltage phasors $\{\hat{V}_i, \hat{\theta}_i\}$ back into the AC power flow Equations (1). Since PQ, PV, and slack buses have different specified and unknown quantities in conventional AC power flow, the residual is computed in a bus-type-aware manner. In particular, the residual is evaluated only for power injections that are specified by the power flow problem formulation, rather than for quantities obtained as dependent variables from the numerical power flow solution. Let $\mathcal{N}_{PQ}$, $\mathcal{N}_{PV}$, and $\mathcal{N}_{ref}$ denote the sets of PQ buses, PV buses, and the slack bus, respectively.

Let P_i^{sp} and Q_i^{sp} denote the specified net active- and reactive-power injections at bus i, respectively, under the sign convention that generation is positive and demand is negative. The specified active-power injection is constructed directly from the input features as

$$P_i^{\mathrm{sp}} = P_i^{\mathrm{gen}} + P_i^{\mathrm{ren}} - P_i^{\mathrm{load}}. \tag{9}$$

For the PQ buses considered in the present datasets, no controllable reactive-power generation is specified. Therefore, the specified reactive-power injection is given by

$$Q_i^{\mathrm{sp}} = -Q_i^{\mathrm{load}}, \quad i \in \mathcal{N}_{PQ}. \tag{10}$$

Fixed shunt capacitors and reactors at PQ buses are not included in Q_i^{sp}. Instead, their shunt susceptances are embedded in the diagonal entries of the scenario-specific bus-admittance matrix and are therefore automatically included in the calculated reactive-power injection $\hat{Q}_i$. This treatment is consistent with the standard AC power flow formulation, in which fixed shunt devices are modeled as network elements rather than controllable reactive-power injections, and avoids double-counting their reactive-power contributions. Thus, P_i^{sp} and Q_i^{sp} are constructed directly from the specified generation and load data and do not use the reactive-power solutions at PV buses or the active- and reactive-power solutions at the slack bus.

The active- and reactive-power injections implied by the predicted voltage solution are calculated as

$$\begin{aligned} \hat{P}_i &= \hat{V}_i \sum_{j \in \overline{\mathcal{N}}_i} \hat{V}_j \big(G_{ij} \cos \hat{\theta}_{ij} + B_{ij} \sin \hat{\theta}_{ij} \big), \\ \hat{Q}_i &= \hat{V}_i \sum_{j \in \overline{\mathcal{N}}_i} \hat{V}_j \big(G_{ij} \sin \hat{\theta}_{ij} - B_{ij} \cos \hat{\theta}_{ij} \big), \end{aligned} \tag{11}$$

where $\hat{\theta}_{ij} = \hat{\theta}_i - \hat{\theta}_j$. Here, $\hat{P}_i$ and $\hat{Q}_i$ denote the active- and reactive-power injections implied by the predicted voltage phasors.

For PQ buses, both active- and reactive-power injections are specified. Therefore, both active- and reactive-power mismatches are included in the physics-informed loss

$$\Delta P_i = P_i^{\mathrm{sp}} - \hat{P}_i, \qquad \Delta Q_i = Q_i^{\mathrm{sp}} - \hat{Q}_i, \quad i \in \mathcal{N}_{PQ}. \tag{12}$$

For PV buses, the active-power injection is specified, whereas the reactive-power injection is a dependent variable determined by the AC power flow equations. Therefore, only the active-power mismatch is included for PV buses

$$\Delta P_i = P_i^{\mathrm{sp}} - \hat{P}_i, \quad i \in \mathcal{N}_{PV}. \tag{13}$$

The reactive-power mismatch is not evaluated at PV buses because Q_i^{sp} is neither constructed nor used for these buses. Consequently, the solved PV-bus reactive-power injection is not introduced into the physics-informed loss. For the slack bus, the voltage magnitude and phase angle are specified as the system reference, whereas the active- and reactive-power injections are dependent variables determined by system power balance and network losses. Therefore, the slack-bus active- and reactive-power mismatches are excluded from the physics residual. The slack-bus voltage magnitude and phase angle remain included in the supervised voltage-output loss for consistency with the full-bus voltage prediction task.

The aggregate physics residual is then defined as

$$\mathcal{L}_{phys} = \frac{1}{|\mathcal{N}_{PQ}|} \sum_{i \in \mathcal{N}_{PQ}} \left(\Delta P_i^2 + \Delta Q_i^2 \right) + \frac{1}{|\mathcal{N}_{PV}|} \sum_{i \in \mathcal{N}_{PV}} \Delta P_i^2. \tag{14}$$

The quantities G_{ij} and B_{ij} used in (11) are obtained from the scenario-specific bus-admittance matrix. Therefore, branch series admittances, line-charging susceptances, transformer tap ratios, bus shunt admittances, fixed shunt capacitors and reactors, and branch-status changes under N$-$1 contingencies are reflected in the residual calculation. In particular, fixed shunt elements are represented through the diagonal entries of the bus-admittance matrix and are included in the calculated injections $\hat{P}_i$ and $\hat{Q}_i$, rather than in the specified injections P_i^{sp} and Q_i^{sp}.

For the neural-network input, an outaged branch is retained in the graph with its branch-status feature set to zero in order to preserve a fixed graph structure. For physics-residual evaluation, the corresponding post-contingency bus-admittance matrix is used so that the electrical contribution of the outaged branch is removed. Generator reactive-power limits are not enforced during data generation. Therefore, Q-limit-induced PV-to-PQ bus-type transitions are not considered in the present study.

This physics-informed regularization is particularly beneficial under distributional shift, which arises naturally in power system operation due to N−1 contingency events and high renewable output variability. Under such conditions, purely data-driven models may encounter operating points that lie outside the training distribution and extrapolate in physically inadmissible directions, producing voltage predictions that appear numerically reasonable but violate nodal power balance. The physics residual penalty provides a corrective signal that reduces violations of the AC power flow equations, improving generalization without requiring additional labeled training data. This property is especially important for dispatch-support and contingency-screening applications under high-renewable-penetration operating conditions, where physically inconsistent power flow predictions may lead to misleading assessments of voltage feasibility, power-balance violations, and operational security margins.

4. Experimental Evaluation

This section presents a comprehensive evaluation of the proposed PI-GAT framework against GCN and GAT baselines on the IEEE 30-bus and IEEE 118-bus benchmark systems. All models are evaluated using voltage prediction errors, phase-angle errors, active/reactive power-mismatch residuals, and load-normalized mismatch metrics. Specifically, **V MAE** and V RMSE measure voltage magnitude prediction accuracy, θ MAE measures phase-angle prediction accuracy, P mis. and Q mis. evaluate active- and reactive-power balance residuals, and P mis./active load and Q mis./active load normalize the corresponding mismatch values by the total system load. The normalized metrics are reported in percentage and are included to facilitate clearer comparison across systems with different load levels and network scales.

4.1. Experimental Setup

Operating scenarios are generated using the pandapower simulation package. The IEEE 30-bus and IEEE 118-bus systems are constructed from the standard pandapower test cases, namely `pn.case30()` and `pn.case118()`. The AC power flow labels are obtained using the pandapower numerical power flow solver (Newton–Raphson) with a convergence tolerance of 1×10^{-8} MVA and a maximum iteration number of 30. Scenarios for which the numerical solver fails to converge are discarded before dataset splitting.

To reflect operating conditions with high renewable penetration, 60% of the eligible PV generation buses, excluding the slack bus, are designated as renewable generation buses. Specifically, using one-based bus numbering, buses 13, 22, and 27 are selected in the IEEE 30-bus system, while buses 1, 6, 8, 12, 18, 19, 25, 27, 31, 34, 40, 42, 49, 55, 56, 61, 65, 70, 72, 74, 77, 80, 87, 90, 91, 99, 103, 104, 107, 111, 112, and 116 are selected in the IEEE 118-bus system. The selected buses remain fixed throughout data generation and are identical for all compared models.

The aggregate rated renewable output is set to 60% of the total base load. In the following, $\mathcal{U}(a, b)$ denotes a uniform distribution on $[a, b]$. The base total active loads of the IEEE 30-bus and IEEE 118-bus systems are approximately 189 MW and 4242 MW, respectively. Individual active loads are independently scaled by factors sampled from $\mathcal{U}(0.80, 1.20)$, while reactive loads are scaled by factors sampled from $\mathcal{U}(0.85, 1.15)$. Renewable source outputs are perturbed as $P_i^{\text{ren}} = P_i^{\text{ren,base}} \cdot \mathcal{U}(0.65, 1.35)$, reflecting the wide variability of renewable generation. To maintain physical plausibility, a curtailment constraint caps the aggregate renewable share at 95% of the current total load. Conventional generator outputs are scaled proportionally to balance the remaining load, with small perturbations $U(0.95, 1.05)$ applied to introduce dispatch variability.

N−1 contingency scenarios are generated by taking one branch out of service at a time and resolving the AC power flow. Each candidate outage is first checked to ensure that it does not create an islanded network. For non-islanding contingencies, the AC power flow is solved using pandapower. Only cases that converge successfully under the specified tolerance are retained, while islanding and non-convergent cases are discarded. The same renewable-bus configuration and contingency-generation procedure are applied to all models. The final dataset contains 75% normal operating scenarios and 25% N−1 contingency scenarios. Before applying the 70/15/15% train/validation/test split, the number of valid scenarios is 6000 and 20,000 for the IEEE 30-bus and IEEE 118-bus systems, respectively. The corresponding numbers of training, validation, and test samples, together with the main model hyperparameters, are summarized in Table 2.

Table 2. Dataset and hyperparameter configuration.

Item	IEEE 30-Bus	IEEE 118-Bus
Total scenarios	6000	20,000
Training samples	4200	14,000
Validation samples	900	3000
Test samples	900	3000
Split (train/val/test)	70/15/15%	70/15/15%
N−1 ratio	25%	25%
Random seed (single run)	2026	2026
Multi-seed experiments	[2026, 2027, 2028, 2029, 2030]	
Epochs	300	500
Batch size	64	64
Learning rate	5×10^{-4}	4×10^{-4}
LR scheduler type	ReduceLROnPlateau	ReduceLROnPlateau
LR scheduler factor	0.5	0.5
LR scheduler patience	8 epochs	25 epochs
Minimum learning rate	1×10^{-6}	1×10^{-6}
Weight decay	1×10^{-5}	1×10^{-5}
Hidden dimension	128	256
GAT layers	3	4
Attention heads	4	4
Dropout	0.1	0.1
λ (physics weight)	0.003	0.0005
Physics warm-up epochs	20	300
Physics ramp epochs	50	50
Early stopping patience	40	70
Early stopping criterion	5×10^{-8}	1×10^{-7}

To benchmark the proposed PI-GAT, two baseline models are considered. The first is a GCN with uniform neighborhood aggregation. The second is an edge-aware GAT that shares the same attention architecture as PI-GAT but sets the physics-loss weight to $\lambda = 0$. For a fair comparison, all models are trained and evaluated using the same generated scenarios, train/validation/test split, input and output definitions, and evaluation metrics. The hidden dimension, number of layers, dropout rate, optimizer settings, batch size, and training epochs are kept comparable across different models. Therefore, the main differences among the compared models lie in the neighborhood aggregation mechanism, the use of edge-aware attention, and the inclusion of the physics-informed residual loss.

As shown in Table 2, different hyperparameter settings are adopted for the IEEE 30-bus and IEEE 118-bus systems to account for their different network scales and training difficulties. In particular, the IEEE 118-bus system uses a larger hidden dimension and a longer physics-loss warm-up period, since the larger network involves more complex electrical interactions and makes early physics-informed optimization less stable.

All models are implemented in Python 3.12.12 using PyTorch 2.9.1+cu128. The experiments are conducted on a workstation equipped with an Intel Core Ultra 9 285K CPU and an NVIDIA RTX 5090 GPU. GPU acceleration is performed using CUDA 12.8. The software environment also includes NumPy 2.3.5 and pandapower 3.2.1.

For model training, the Adam optimizer is used with $\beta_1 = 0.9$, $\beta_2 = 0.999$, and a weight decay of 1×10^{-5}. Each of the five independent training runs uses a different random seed, which simultaneously varies the train/validation/test split, the model weight initialization, and the mini-batch shuffle order. Gradient clipping with a maximum norm of 2.0 is applied to improve training stability. A ReduceLROnPlateau learning-rate scheduler is adopted, where the validation loss is used as the monitored metric. The detailed learning rates, scheduler patience values, and other training hyperparameters are provided in Table 2.

For PI-GAT, the physics loss weight λ is introduced through a warm-up-ramp schedule instead of being applied from the first training epoch. Specifically, λ is set to zero during the warm-up period and is then linearly increased to its target value during the ramp period. This design allows the supervised regression loss to dominate the early stage of training, while the AC power flow residual gradually contributes to the gradient updates in later epochs.

4.2. Results on IEEE 30-Bus System

The IEEE 30-bus system (Figure 2) serves as the first benchmark, comprising 30 buses and 41 branches. Figure 3 shows the training dynamics of PI-GAT on this system. Both the total loss and the physics residual exhibit generally decreasing and stable trends. This continuous convergence indicates that the short warm-up period is sufficient for a network of this scale, allowing the physics constraints to be introduced into an already well-conditioned optimization landscape. Consequently, the model achieves stable voltage predictions while improving physical consistency and reducing the power-balance deviations commonly observed in purely data-driven approaches.

Table 3 presents the IEEE 30-bus test-set performance over five independent runs. Compared with GCN, GAT improves voltage and angle prediction accuracy, indicating the benefit of adaptive attention-based neighborhood aggregation. PI-GAT achieves a voltage MAE comparable to GAT, with mean values of 0.000984 and 0.000989, respectively, while introducing a modest increase in voltage RMSE and angle MAE.

Table 3. Performance comparison on the IEEE 30-bus system over five independent runs.

Model	V MAE	V RMSE	θ MAE (°)	P Mis.	Q Mis.	P Mis./Active Load (%)	Q Mis./Active Load (%)
GCN	0.001170 (2.8×10^{-5})	0.003335 (1.0×10^{-5})	0.2153 (3.6×10^{-3})	0.0242 (2.2×10^{-4})	0.0130 (3.5×10^{-4})	1.28 (1.1×10^{-2})	0.6854 (1.8×10^{-2})
GAT	0.000989 (4.7×10^{-5})	0.001758 (5.5×10^{-5})	0.1704 (3.1×10^{-3})	0.0187 (5.9×10^{-4})	0.0115 (6.7×10^{-4})	0.9874 (3.1×10^{-2})	0.6063 (3.5×10^{-2})
PI-GAT	0.000984 (4.5×10^{-5})	0.001975 (1.1×10^{-4})	0.1804 (4.4×10^{-3})	0.0139 (2.1×10^{-4})	0.007607 (2.0×10^{-4})	0.7341 (1.1×10^{-2})	0.4020 (1.1×10^{-2})

Note: Values are reported as mean (standard deviation) over five independent runs. V-related errors and power mismatches are reported in p.u.; θ MAE is reported in degrees. P/load and Q/load denote active- and reactive-power mismatches normalized by the total active load.

The main advantage of PI-GAT lies in physical consistency. Compared with GAT without the physics-informed loss, PI-GAT reduces the mean active-power mismatch from 0.0187 to 0.0139 and the mean reactive-power mismatch from 0.0115 to 0.007607. In normalized terms, P/load decreases from 0.9874% to 0.7341%, and Q/load decreases from 0.6063% to 0.4020%. These results indicate that PI-GAT improves power-balance consistency while maintaining voltage prediction accuracy at the same order of magnitude as the data-driven GAT baseline. The small standard deviations further show that these trends are stable across five independent runs.

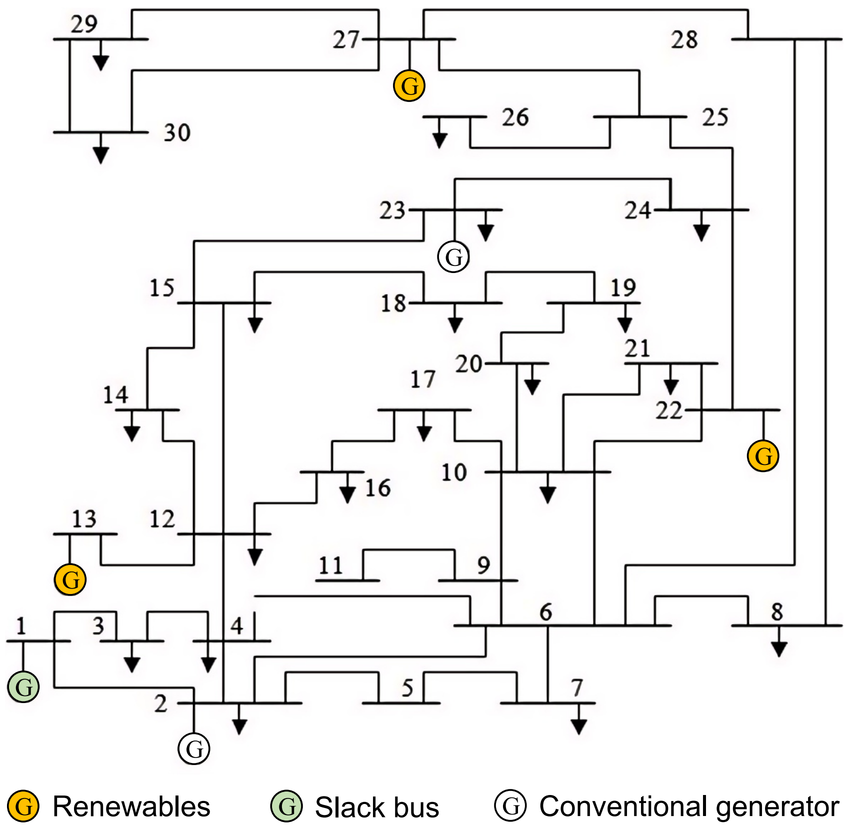


Figure 2. IEEE 30-bus system.

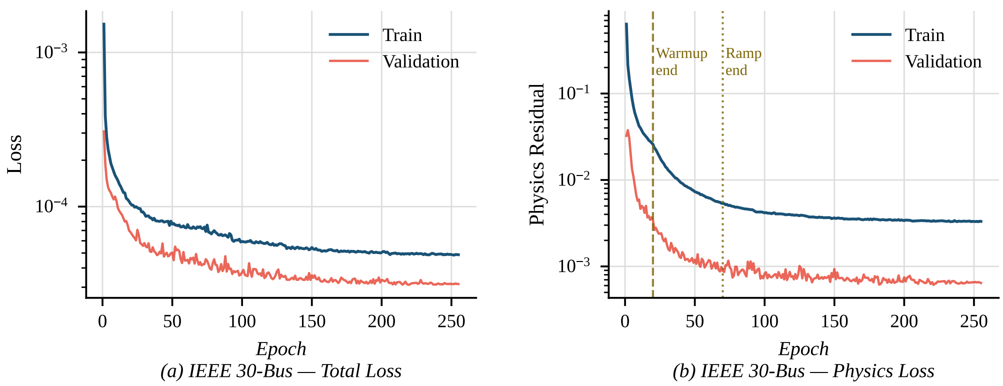


Figure 3. Training and validation loss curves of PI-GAT on the IEEE 30-bus system. (**a**) total loss; (**b**) physics residual loss.

To further evaluate scenario-separated generalization, the IEEE 30-bus test set is divided into normal operating samples and N−1 contingency samples, as reported in Table 4. Under normal operating conditions, PI-GAT achieves the lowest voltage MAE and produces the smallest active- and reactive-power mismatches among the three models. Compared with GAT without the physics-informed loss, PI-GAT reduces P/load from 0.8806% to 0.6459% and Q/load from 0.5361% to 0.3585%. This indicates that the physics-informed residual improves physical consistency without degrading voltage magnitude accuracy under normal operating conditions.

Table 4. Generalization performance on the IEEE 30-bus system under normal and N−1 contingency conditions over five independent runs.

Condition	Model	V MAE	V RMSE	θ MAE (°)	P Mis.	Q Mis.	P Mis./Active Load (%)	Q Mis./Active Load (%)
Normal	GCN	0.0008 (2.8×10^{-5})	0.0011 (3.7×10^{-5})	0.1594 (4.4×10^{-3})	0.0245 (2.2×10^{-4})	0.0130 (4.2×10^{-4})	1.2938 (1.2×10^{-2})	0.6894 (2.2×10^{-2})
Normal	GAT	0.0008 (3.9×10^{-5})	0.0009 (3.0×10^{-5})	0.1369 (1.9×10^{-3})	0.0167 (7.0×10^{-4})	0.0101 (8.4×10^{-4})	0.8806 (3.7×10^{-2})	0.5361 (4.4×10^{-2})
Normal	PI-GAT	0.0007 (2.6×10^{-5})	0.0009 (2.4×10^{-5})	0.1472 (4.6×10^{-3})	0.0122 (3.4×10^{-4})	0.0068 (1.2×10^{-4})	0.6459 (1.8×10^{-2})	0.3585 (6.1×10^{-3})
N−1	GCN	0.0023 (3.7×10^{-5})	0.0061 (2.6×10^{-5})	0.4006 (2.2×10^{-3})	0.0272 (3.0×10^{-4})	0.0150 (4.6×10^{-4})	1.4375 (1.6×10^{-2})	0.7952 (2.4×10^{-2})
N−1	GAT	0.0017 (8.3×10^{-5})	0.0031 (1.4×10^{-4})	0.3035 (7.1×10^{-3})	0.0259 (1.4×10^{-3})	0.0159 (8.5×10^{-4})	1.3711 (7.5×10^{-2})	0.8422 (4.5×10^{-2})
N−1	PI-GAT	0.0017 (9.4×10^{-5})	0.0035 (2.2×10^{-4})	0.3230 (9.6×10^{-3})	0.0182 (3.5×10^{-4})	0.0105 (2.4×10^{-4})	0.9620 (1.9×10^{-2})	0.5568 (1.3×10^{-2})

Note: Values are reported as mean (standard deviation) over five independent runs. Normal denotes test samples without line outages, while N−1 denotes test samples with single-line outages. V-related errors and power mismatches are reported in p.u.; θ MAE is reported in degrees.

Under N−1 contingency conditions, all models exhibit larger voltage and angle errors than in the normal case, reflecting the additional difficulty introduced by line outages and topology perturbations. In this setting, GAT achieves slightly lower voltage RMSE and angle MAE than PI-GAT. However, PI-GAT maintains better physical consistency, reducing P/load from 1.3711% to 0.9620% and Q/load from 0.8422% to 0.5568% compared with GAT. These results show that the physics-informed residual substantially improves normalized power-balance consistency under post-contingency conditions, although it may introduce a slight trade-off in point-wise voltage or angle accuracy.

4.3. Results on IEEE 118-Bus System

The IEEE 118-bus system (Figure 4) provides a larger-scale benchmark with 118 buses and 186 branches. Due to the increased topological complexity, the physics loss warm-up period is extended to 300 epochs. This ensures that the regression objective first establishes a stable baseline approximation of the system state before the physics-informed regularization is applied. As shown in Figure 5, the total loss exhibits a distinct inflection point at epoch 300, accompanied by a sharp drop in the physics residual. This behavior suggests that the delayed activation strategy effectively stabilizes training for larger networks, successfully guiding the model from an initial data-driven fit toward highly accurate voltage predictions with improved physical consistency of voltage magnitude and phase angle.

Table 5 reports the IEEE 118-bus results over five independent runs. Compared with the IEEE 30-bus system, the larger network contains more buses, branches, and generator buses, making accurate and physically consistent prediction more challenging. GCN and GAT show comparable voltage prediction performance on this system: GCN obtains the lowest voltage MAE, while GAT achieves the lowest voltage RMSE and angle MAE. This indicates that the data-driven baselines can provide accurate point-wise voltage predictions, but their physical consistency remains limited.

PI-GAT shows a clear advantage in reducing power-balance residuals. Compared with GAT without the physics-informed loss, PI-GAT reduces the mean active-power mismatch from 0.6043 to 0.2319 and the mean reactive-power mismatch from 0.1383 to 0.0525. In normalized terms, P/load decreases from 1.42% to 0.5466%, and Q/load decreases from 0.3261% to 0.1237%. This confirms that the main benefit of PI-GAT on the larger IEEE

118-bus system lies in improving physical consistency, although this improvement is accompanied by larger point-wise voltage and angle errors than the purely data-driven GAT baseline. The small standard deviations indicate that this trade-off and the physical-consistency improvement are stable across five independent runs.

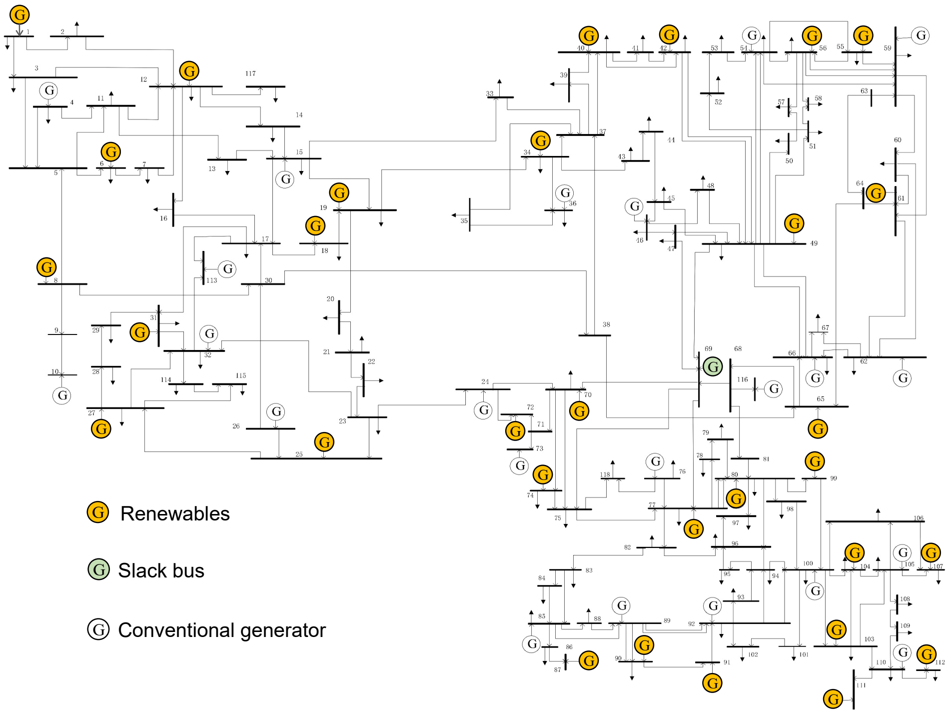
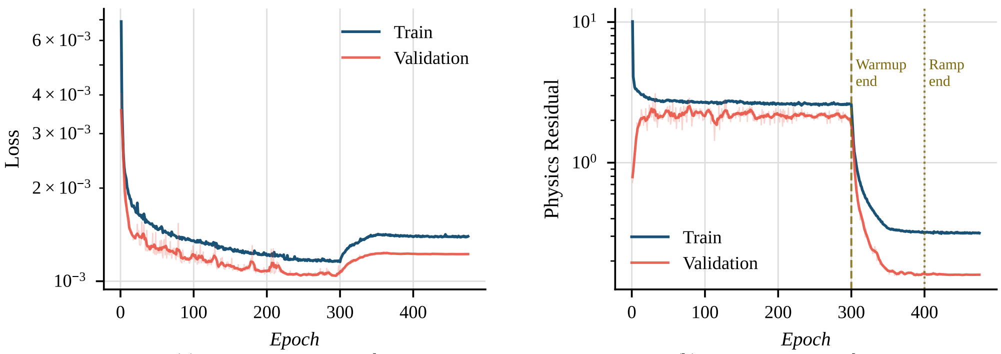


Figure 4. IEEE 118-bus system.



(a) IEEE 118-Bus — Total Loss *(b) IEEE 118-Bus — Physics Loss*

Figure 5. Training and validation loss curves of PI-GAT on the IEEE 118-bus system. (**a**) total loss; (**b**) physics residual loss.

Table 5. Performance comparison on the IEEE 118-bus system over five independent runs.

Model	V MAE	V RMSE	θ MAE (°)	P Mis.	Q Mis.	P Mis./Active Load (%)	Q Mis./Active Load (%)
GCN	0.000614 (3.1×10^{-5})	0.001151 (2.2×10^{-5})	1.16 (9.6×10^{-3})	0.6427 (1.0×10^{-2})	0.1421 (2.3×10^{-3})	1.51 (2.5×10^{-2})	0.3350 (5.5×10^{-3})
GAT	0.000669 (1.3×10^{-5})	0.001075 (8.4×10^{-6})	1.12 (5.7×10^{-3})	0.6043 (5.5×10^{-3})	0.1383 (1.7×10^{-3})	1.42 (1.3×10^{-2})	0.3261 (3.9×10^{-3})
PI-GAT	0.000934 (1.2×10^{-5})	0.001505 (2.3×10^{-5})	1.21 (7.6×10^{-3})	0.2319 (4.2×10^{-3})	0.0525 (1.7×10^{-3})	0.5466 (9.8×10^{-3})	0.1237 (4.0×10^{-3})

Note: Values are reported as mean (standard deviation) over five independent runs. V-related errors and power mismatches are reported in p.u.; θ MAE is reported in degrees. P/load and Q/load denote active- and reactive-power mismatches normalized by the total active load.

Table 6 further reports the scenario-separated generalization performance on the IEEE 118-bus system under normal operating and N−1 contingency conditions. Under normal operating conditions, GCN and GAT obtain lower voltage and angle errors than PI-GAT. Nevertheless, PI-GAT substantially reduces the active- and reactive-power mismatches. Compared with GAT without the physics-informed loss, PI-GAT reduces P/load from 1.4327% to 0.5451% and Q/load from 0.3282% to 0.1235%. This demonstrates that the embedded AC power flow residual guides the model toward solutions that better satisfy nodal power balance, even when the supervised voltage error is not the lowest.

Table 6. Generalization performance on the IEEE 118-bus system under normal and N−1 contingency conditions over five independent runs.

Condition	Model	V MAE	V RMSE	θ MAE (°)	P Mis.	Q Mis.	P Mis./Active Load (%)	Q Mis./Active Load (%)
Normal	GCN	0.0006 (3.2×10^{-5})	0.0008 (3.2×10^{-5})	1.0867 (1.0×10^{-2})	0.6509 (1.0×10^{-2})	0.1447 (2.5×10^{-3})	1.5335 (2.4×10^{-2})	0.3410 (5.8×10^{-3})
Normal	GAT	0.0006 (1.4×10^{-5})	0.0009 (1.9×10^{-5})	1.0729 (2.1×10^{-3})	0.6081 (5.3×10^{-3})	0.1393 (1.7×10^{-3})	1.4327 (1.3×10^{-2})	0.3282 (4.1×10^{-3})
Normal	PI-GAT	0.0009 (1.1×10^{-5})	0.0013 (2.4×10^{-5})	1.1595 (4.1×10^{-3})	0.2314 (3.8×10^{-3})	0.0524 (1.7×10^{-3})	0.5451 (8.9×10^{-3})	0.1235 (4.0×10^{-3})
N−1	GCN	0.0007 (3.1×10^{-5})	0.0017 (1.6×10^{-5})	1.2038 (8.0×10^{-3})	0.6462 (1.1×10^{-2})	0.1443 (2.3×10^{-3})	1.5228 (2.6×10^{-2})	0.3399 (5.5×10^{-3})
N−1	GAT	0.0007 (1.3×10^{-5})	0.0014 (7.3×10^{-6})	1.1625 (4.3×10^{-3})	0.6076 (4.4×10^{-3})	0.1396 (1.6×10^{-3})	1.4318 (1.0×10^{-2})	0.3289 (3.7×10^{-3})
N−1	PI-GAT	0.0010 (1.1×10^{-5})	0.0017 (1.8×10^{-5})	1.2435 (5.9×10^{-3})	0.2364 (3.8×10^{-3})	0.0542 (1.6×10^{-3})	0.5570 (9.0×10^{-3})	0.1277 (3.8×10^{-3})

Note: Values are reported as mean (standard deviation) over five independent runs. Normal denotes test samples without line outages, while N−1 denotes test samples with single-line outages. V-related errors and power mismatches are reported in p.u.; θ MAE is reported in degrees.

A similar trend is observed under N−1 contingency conditions. Although PI-GAT has slightly higher voltage and angle errors than the purely data-driven GAT baseline, it consistently yields much smaller normalized power-mismatch values. Specifically, PI-GAT reduces P/load from 1.4318% to 0.5570% and Q/load from 0.3289% to 0.1277% compared with GAT. The preservation of this advantage under contingency scenarios indicates that the physics-informed residual remains effective for post-contingency samples included in the test distribution. Therefore, for the larger IEEE 118-bus system, PI-GAT should be interpreted as providing a more physically consistent surrogate rather than uniformly minimizing every point-wise voltage or angle error.

4.4. Application-Oriented Evaluation

To further validate the applicability of predicted voltage states for operational evaluation, we adopt application-oriented metrics on the IEEE 118-bus test system with 3000 test samples. Metrics are split into two categories. Bus-level metrics include voltage magnitude error, phase angle error, and active/reactive power mismatches. Branch-level metrics cover active and reactive line flow errors.

All metrics share a consistent averaging process unless stated otherwise. We first calculate metric values for each test scenario and average the results over all samples. For bus-level MAE, absolute residuals are averaged over relevant buses in every sample: non-slack buses for active power and only PQ buses for reactive power. Per-sample averages are then aggregated to obtain the final MAE.

The 95th-percentile error uses the same two-step workflow. We first extract the 95th percentile of bus-wise residuals within each sample, then average these percentile values across the test set. The maximum bus-level error is the largest absolute residual observed over all buses and test samples.

Branch flow errors follow identical averaging rules for MAE and the 95th percentile. All branch-flow quantities are evaluated at the from-bus end, as defined by the branch orientation in the MATPOWER data structure. These metrics directly quantify how surrogate predictions influence power-flow-based security analysis.

As shown in Table 7, PI-GAT achieves a voltage-magnitude MAE of 0.000909 p.u. and a 95th-percentile voltage-magnitude error of 0.002554 p.u., indicating that most voltage-magnitude predictions remain accurate. At the branch level, the active and reactive branch-flow MAEs are 13.2261 MW and 3.5935 MVAr, respectively. These results suggest that PI-GAT preserves voltage magnitudes and branch power flow information with reasonable accuracy for most test samples.

Table 7. Application-oriented evaluation metrics on the IEEE 118-bus test set.

Metric	MAE	Mean Per-Scenario P95	Maximum
Voltage magnitude error (p.u.)	0.000909	0.002554	0.071537
Voltage angle error (°)	1.2025	3.8126	16.9823
Active-power mismatch (MW)	25.0495	87.9206	608.0801
Reactive-power mismatch (MVAr)	5.7410	18.0986	129.9013
Active branch-flow error (MW)	13.2261	56.7225	146.5554
Reactive branch-flow error (MVAr)	3.5935	14.6191	36.7699

Note: This table reports the test outcomes corresponding to random seed 2026.

The maximum errors are larger than the average and 95th-percentile values, indicating the presence of a small number of difficult post-contingency samples. This is expected because certain N−1 outages can cause stronger redistribution of power flows and voltage phase angles than typical samples. Therefore, the maximum-error values should be interpreted as worst-case stress indicators rather than as representative errors of the overall test distribution. For critical operating points or near-limit contingencies, the surrogate should be used for fast screening and should be followed by conventional AC power flow verification before final operational decisions are made.

4.5. Ablation Study

To identify the contribution of each component in PI-GAT, an ablation study is conducted on the IEEE 118-bus system. The main performance comparison has already been reported on both the IEEE 30-bus and IEEE 118-bus systems. The ablation study is performed on the IEEE 118-bus system because it is larger and contains more buses, branches, generators, and possible N−1 contingency scenarios. Therefore, it provides a more challenging and representative setting for evaluating the effects of edge-aware attention, physics-informed residual regularization, branch-status contingency encoding, and warm-up-ramp scheduling.

As shown in Table 8, the physics-informed residual is the main factor responsible for reducing active- and reactive-power mismatches. Compared with GAT without the physics-informed loss, the full PI-GAT reduces the active-power mismatch from 0.6119 to 0.2585 and the reactive-power mismatch from 0.1382 to 0.0597, corresponding to reductions of approximately 57.8% and 56.8%, respectively. The comparison between node-only GAT with the physics-informed loss and the full PI-GAT further indicates that physics-informed regularization alone can already improve physical consistency, while edge-aware attention helps maintain a more balanced trade-off between voltage prediction and physical residuals.

The branch-status ablation further shows the importance of explicitly encoding contingency information. Removing branch-status features increases the voltage MAE from 0.000980 to 0.001074 and increases the active- and reactive-power mismatches from 0.2585 and 0.0597 to 0.2755 and 0.0636, respectively. This confirms that branch-status features help the model distinguish post-contingency operating conditions within the fixed-graph representation.

Table 8. Ablation study on the IEEE 118-bus N−1 contingency test set.

Model	Voltage MAE	Voltage RMSE	Angle MAE (°)	P Mismatch	Q Mismatch
GCN	0.000744	0.001747	1.1991	0.6461	0.1428
GAT without physics-informed loss	0.000739	0.001405	1.1667	0.6119	0.1382
Node-only GAT with physics-informed loss	0.000976	0.001961	1.2776	0.2381	0.0559
PI-GAT without warm-up-ramp scheduling	0.001030	0.001832	1.2659	0.2297	0.0508
PI-GAT without branch-status features	0.001074	0.002205	1.3022	0.2755	0.0636
Full PI-GAT	0.000980	0.001668	1.2371	0.2585	0.0597

Note: This table reports the test outcomes corresponding to random seed 2026.

The comparison between the full PI-GAT and PI-GAT without warm-up-ramp scheduling shows that directly applying the physics-informed loss from the beginning of training can further reduce the mismatch residuals in this case, but it also increases voltage and angle errors. Therefore, the warm-up-ramp strategy is retained because it provides a more balanced optimization process between supervised voltage fitting and physical consistency.

4.6. Sensitivity to the Physics-Loss Weight

To further examine the effect of the physics-informed loss weight, a sensitivity analysis is conducted on the IEEE 118-bus N−1 contingency test set. The purpose of this analysis is to evaluate how λ affects the trade-off between point-wise voltage prediction accuracy and physical consistency. Since the supervised voltage-regression loss and the AC residual loss have different numerical scales, a small numerical value of λ does not imply that the physical constraint is unimportant. Instead, λ balances the gradient contributions of the data-fitting term and the physics-residual term during training.

As shown in Table 9, increasing λ generally reduces active- and reactive-power mismatches, but it also increases voltage and angle prediction errors. When $\lambda = 10^{-4}$, the model achieves the lowest voltage MAE among the tested settings, but the active- and reactive-power mismatches remain relatively large. Increasing λ to 10^{-3} and 5×10^{-3} further improves physical consistency, but the voltage MAE and phase-angle MAE become noticeably larger. Therefore, the selected setting is not chosen to minimize only the residual mismatch. Instead, it is selected as a balanced value that substantially improves physical consistency while avoiding excessive degradation of point-wise voltage and angle accuracy.

Table 9. Sensitivity analysis of the physics-loss weight on the IEEE 118-bus N−1 contingency test set.

Model	V MAE	V RMSE	θ MAE (°)	P Mis.	Q Mis.
PI-GAT, $\lambda = 10^{-4}$	0.000819	0.001471	1.1990	0.3522	0.0772
PI-GAT, $\boldsymbol{\lambda = 5 \times 10^{-4}}$	0.000980	0.001668	1.2371	0.2585	0.0597
PI-GAT, $\lambda = 10^{-3}$	0.001119	0.001890	1.2935	0.1856	0.0436
PI-GAT, $\lambda = 5 \times 10^{-3}$	0.001451	0.002305	1.4201	0.1288	0.0344

Note: This table reports the test outcomes corresponding to random seed 2026. The bolded row represents the hyperparameter configuration adopted for all main experiments in this paper.

4.7. Generalization to Unseen N−1 Contingencies

To further evaluate whether PI-GAT can generalize to topology changes not observed during training, an unseen N−1 contingency split is constructed on the IEEE 118-bus system. Among the 166 unique N−1 line-outage types present in the IEEE 118 dataset, 116 lines (70%) were allocated to training, 24 (14.5%) to validation, and 26 (15.6%) were entirely withheld for testing. Normal no-outage samples were split randomly across the three sets in the standard 70/15/15 proportion. No training or validation sample contains an outage on any of these 26 lines, ensuring that the test evaluates generalization to genuinely unseen topological configurations. The metrics in Table 10 are calculated exclusively over the N−1 samples associated with the 26 withheld outage types.

Table 10. Results on unseen IEEE 118-bus N−1 contingencies.

Model and Split	Voltage MAE	Voltage RMSE	Angle MAE (°)	P Mismatch	Q Mismatch
GAT without physics-informed loss, standard split	0.000731	0.001247	1.1256	0.5957	0.1331
PI-GAT, standard split	0.000963	0.001514	1.2073	0.2504	0.0573
GAT without physics-informed loss, unseen N−1 split	0.000788	0.001321	1.1531	0.6004	0.1373
PI-GAT, unseen N−1 split	0.001006	0.001566	1.2392	0.2431	0.0593

Note: This table reports the test outcomes corresponding to random seed 2026.

As shown in Table 10, the unseen-contingency split is more stringent than the standard random split because the tested line-outage types are not observed during training. Under this setting, PI-GAT reduces the active-power mismatch from 0.6004 to 0.2431 and the reactive-power mismatch from 0.1373 to 0.0593 compared with GAT without the physics-informed loss, corresponding to reductions of approximately 59.5% and 56.8%, respectively. These reductions are close to those observed under the standard split, indicating that the physics-informed residual improves physical consistency not only for randomly mixed contingency samples but also for unseen outage types.

Although PI-GAT has slightly larger voltage and angle errors than the purely data-driven GAT baseline, it provides substantially smaller power-balance residuals, which is important for physically reliable contingency screening.

4.8. Noise Sensitivity Analysis

To examine the robustness of the proposed PI-GAT model under measurement uncertainty, a noise sensitivity analysis is conducted on both the IEEE 30-bus and IEEE 118-bus test sets. During testing, independent zero-mean Gaussian noise is added to the continuous measured input features $x \in P^{\text{load}}, Q^{\text{load}}, P^{\text{ren}}, P^{\text{gen}}$, with standard deviation $\sigma = \eta|x|$, where $\eta \in \{0, 0.5\%, 1\%, 2\%, 5\%\}$. Voltage setpoints, bus-type indicators, and branch-status variables are kept unchanged because they represent scheduled or categorical information rather than measured continuous quantities. The noise standard deviation is set as a percentage of the corresponding feature magnitude. The model is trained on the original noise-free training set and is directly evaluated under noisy test inputs, so this experiment measures the robustness of the trained model to measurement noise rather than retraining adaptation.

As shown in Table 11, PI-GAT exhibits stable performance under moderate measurement noise. For the IEEE 30-bus system, increasing the noise level from 0% to 2% only slightly changes the voltage MAE from 0.000956 to 0.000964 and the P mismatch from 1.40 to 1.44. A larger noise level of 5% leads to more visible degradation, with the voltage MAE increasing to 0.001004 and the P mismatch increasing to 1.69. A similar trend is observed on the IEEE 118-bus system. The voltage MAE increases from 0.000919 under noise-free inputs to 0.000924 under 2% noise and 0.000944 under 5% noise, while the P mismatch increases from 25.26 to 25.70 and 27.72, respectively.

Table 11. Noise-sensitivity results of PI-GAT on the IEEE 30-bus and IEEE 118-bus systems under Gaussian noise injected into continuous input features.

System	Noise Level	V MAE	V RMSE	θ MAE (°)	P Mismatch (MW)	Q Mismatch (MVAr)
IEEE 30	0%	0.000956	0.001905	0.1762	1.40	0.74
IEEE 30	0.5%	0.000957	0.001905	0.1771	1.40	0.74
IEEE 30	1%	0.000957	0.001905	0.1782	1.41	0.74
IEEE 30	2%	0.000964	0.001910	0.1847	1.44	0.75
IEEE 30	5%	0.001004	0.001948	0.2262	1.69	0.81
IEEE 118	0%	0.000919	0.001483	1.2158	25.26	5.77
IEEE 118	0.5%	0.000919	0.001483	1.2196	25.30	5.77
IEEE 118	1%	0.000920	0.001484	1.2274	25.40	5.79
IEEE 118	2%	0.000924	0.001489	1.2721	25.70	5.85
IEEE 118	5%	0.000944	0.001514	1.4932	27.72	6.23

These results indicate that the proposed model has a certain degree of robustness to moderate measurement noise in the input features. However, the degradation under larger noise levels also shows that real-world deployment should consider measurement uncertainty explicitly. Because all training and evaluation data are simulation-generated, the present results demonstrate performance under controlled benchmark conditions but do not eliminate the synthetic-to-real domain gap. Validation using SCADA, PMU, or state-estimation data remains necessary before field deployment.

4.9. Computational Efficiency

To provide a fairer computational-efficiency comparison, we redesign the timing protocol and separately evaluate numerical power flow solvers and PI-GAT inference under different device and batching settings. All numerical solvers are executed using pandapower with the PYPOWER backend on CPU only, since pandapower does not support GPU acceleration. The Newton–Raphson solver is configured with a maximum iteration number of 30 and a convergence tolerance of 1×10^{-8} MVA. Voltage-angle calculation is enabled, generator reactive-power limits are not enforced, and the convergence check is based on the MVA mismatch. Only successfully converged cases are included in the timing comparison.

For numerical solvers, only the `pp.runpp()` call is timed, while network copying, scenario perturbation, and input assembly are excluded. For PI-GAT, only the `model.forward()` call is timed. Data transfer between CPU and GPU, feature standardization, and input assembly are excluded. Graph construction, including `edge_index` and edge attributes, is treated as a one-time cost for a fixed network topology and is not included in per-scenario inference time. The timing evaluation is performed on 100 successfully converged test scenarios after 10 warm-up runs. For batched PI-GAT inference, the batch size is set to 64.

As shown in Table 12, PI-GAT provides clear acceleration over Newton–Raphson power flow under both CPU and GPU inference settings. For single-scenario inference, PI-GAT achieves speedups of 16.93× and 8.74× on CPU for the IEEE 30-bus and IEEE 118-bus systems, respectively. GPU single-scenario inference also achieves speedups of 9.59× and 8.12×, respectively. The CPU single-scenario inference is slightly faster than GPU single-scenario inference in these benchmark systems because the graphs are relatively small and GPU kernel-launch overhead can dominate the actual computation time.

Table 12. Computational-efficiency comparison under different solver and inference settings.

System	Method	Device/Mode	Time per Scenario (ms)	Speedup over NR
IEEE 30	Newton–Raphson power flow	CPU, single	10.8434	1.00×
IEEE 30	Fast Decoupled BX power flow	CPU, single	10.9951	0.99×
IEEE 30	Fast Decoupled XB power flow	CPU, single	10.9726	0.99×
IEEE 30	Iwamoto Newton–Raphson	CPU, single	11.2513	0.96×
IEEE 30	PI-GAT	GPU, single	1.1307	9.59×
IEEE 30	PI-GAT	GPU, batch-64	0.0272	398.65×
IEEE 30	PI-GAT	CPU, single	0.6406	16.93×
IEEE 30	PI-GAT	CPU, batch-64	0.0665	163.06×
IEEE 118	Newton–Raphson power flow	CPU, single	12.2187	1.00×
IEEE 118	Fast Decoupled BX power flow	CPU, single	12.0824	1.01×
IEEE 118	Fast Decoupled XB power flow	CPU, single	12.2117	1.00×
IEEE 118	Iwamoto Newton–Raphson	CPU, single	12.5392	0.97×
IEEE 118	PI-GAT	GPU, single	1.5052	8.12×
IEEE 118	PI-GAT	GPU, batch-64	0.0475	257.24×
IEEE 118	PI-GAT	CPU, single	1.3978	8.74×
IEEE 118	PI-GAT	CPU, batch-64	0.4428	27.59×

Note: The speedup is computed relative to the Newton–Raphson power flow time of the same system. For PI-GAT batch-64 inference, the reported value denotes the average per-scenario time within a batch.

The advantage of GPU inference becomes more evident in batched evaluation. With a batch size of 64, PI-GAT reduces the per-scenario inference time to 0.0272 ms on the IEEE 30-bus system and 0.0475 ms on the IEEE 118-bus system, corresponding to speedups of 398.65× and 257.24× over Newton–Raphson power flow, respectively. This result is consistent with the intended use of the surrogate model for repeated multi-scenario assessment, where many renewable-generation, load-variation, and contingency scenarios are evaluated under the same network topology.

It should also be emphasized that the reported PI-GAT timing corresponds to forward-pass inference after the graph structure and model inputs are available. Therefore, the results should be interpreted as inference-time acceleration for repeated power flow assessment, rather than as a full end-to-end replacement of all data-preparation procedures. Since the graph representation is fixed for a given benchmark network and branch outages are encoded through edge-status features, graph construction can be reused across scenarios and is treated as an offline or one-time preparation step.

4.10. Training Time and Scalability Discussion

In addition to online inference efficiency, the offline training time is also evaluated because PI-GAT must be trained before deployment. Table 13 summarizes the training time of GCN, GAT, and PI-GAT on a single GPU. The IEEE 118-bus system requires a longer training time than the IEEE 30-bus system for all compared models because it contains more buses, branches, and operating scenarios, and because larger model dimensions and more training epochs are used for the larger system.

Table 13. Training-time comparison on a single GPU.

Model	IEEE 30-Bus	IEEE 118-Bus
GCN	∼2.6 min	∼16 min
GAT	∼3.6 min	∼34 min
PI-GAT	∼4.4 min	∼37 min

Note: Values are averaged over five independent runs.

As shown in Table 13, PI-GAT requires approximately 4.4 min on the IEEE 30-bus system and approximately 37 min on the IEEE 118-bus system. The increase is mainly due to the larger graph size, larger hidden dimension, more training samples, and longer training schedule used for the IEEE 118-bus system. Compared with GAT without the physics-informed loss, the additional training cost of PI-GAT is moderate on the IEEE 118-bus system, increasing from approximately 34 min to 37 min. This indicates that the physics-informed residual introduces additional computational overhead, but the dominant cost still comes from graph neural-network training and message passing on the larger dataset.

It should also be clarified that the number of trainable parameters in PI-GAT is not directly proportional to the number of buses. The graph attention layers share their weights across all buses and branches. Therefore, the parameter count is mainly determined by the input feature dimension, hidden dimension, number of attention heads, and number of layers, rather than by assigning independent weights to each bus. However, the computational cost per epoch does increase with the number of buses and branches because message passing is performed over more edges and the AC residual is evaluated over more buses.

For systems with thousands of buses, the proposed framework remains conceptually applicable because it is based on shared graph operations and fixed-graph branch-status encoding. Nevertheless, practical deployment on such large-scale systems would require additional engineering considerations, including sparse tensor implementation, efficient batching over operating scenarios, graph partitioning, hierarchical message passing, and optimized physics-residual evaluation. Therefore, the IEEE 30-bus and IEEE 118-bus experiments should be interpreted as benchmark validation of the proposed framework, while scaling to systems with thousands of buses is an important direction for future work.

5. Conclusions

This paper proposed PI-GAT, a physics-informed graph attention network for rapid and physically consistent power flow assessment in power systems with high penetration of renewable generation. By adding a bus-type-aware AC power flow residual to the training objective, PI-GAT substantially reduces nodal power-balance mismatches, which is essential for reliable security assessment under complex operating conditions. The edge-aware multi-head attention mechanism allows the model to adaptively modulate neighbor influence according to branch electrical parameters, enabling physically meaningful information aggregation along the network topology. A warm-up-ramp scheduling strategy is further introduced to stabilize physics-informed training, allowing the regression objective to first establish a stable voltage approximation before the physics constraint begins to shape the gradient.

Experiments on the IEEE 30-bus and IEEE 118-bus benchmark systems demonstrate the effectiveness of the proposed framework. PI-GAT reduces active and reactive power mismatch by up to approximately 62% relative to the edge-aware GAT baseline while maintaining voltage errors within the evaluated tolerance range, with the physical consistency advantage preserved across both normal and $N-1$ contingency scenarios. Furthermore, PI-GAT achieves up to 16.93× speedup in single-scenario inference and up to 398.65× speedup in batched inference under the reported core-inference timing protocol. These results suggest that embedding physics constraints directly into the training objective provides a more reliable path toward more physically consistent surrogate models than purely data-driven approaches, particularly under distributional shift introduced by renewable variability and contingency events. Future work will investigate adaptive physics loss weighting, the extension to unbalanced distribution networks within high renewable contexts, and the integration of PI-GAT within broader dispatch optimization and security-assessment frameworks.

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References

1. Liu, Y.; Wang, Z.; Huang, R.; Yang, B. Nash bargaining for power-fatigue co-optimization in wake-affected wind farms: A learning-aided approach. *Control Eng. Pract.* **2026**, *169*, 106809. [CrossRef]
2. Li, M.; Wang, Z.; Liu, F.; Jia, M.; Yang, B. Distributed stability-guaranteed reinforcement learning for the market-driven frequency control in power systems. *CSEE J. Power Energy Syst.* **2025**. [CrossRef]
3. International Energy Agency. *Integrating Solar and Wind*; Technical Report; IEA: Paris, France, 2024.
4. Ren, Y.; Wang, Z.; Liu, F. Small-signal security region of floating offshore wind turbine under wind and wave disturbances. *IEEE Trans. Sustain. Energy* **2026**. [CrossRef]
5. Wang, Z.; Hao, W.; Wei, W.; Sun, Z. Online Distributed Generalized Nash Equilibrium Seeking of Energy Sharing Markets in Distribution Networks. *IEEE Trans. Autom. Sci. Eng.* **2025**, *22*, 19470–19483. [CrossRef]
6. Wang, W.; Liu, Y.; Wang, Z. Online Voltage Regulation of Distribution Networks in Energy Sharing Markets: A Feedback-Based Method With Parameter Estimation. *IEEE Trans. Smart Grid* **2025**, *16*, 2991–3002. [CrossRef]
7. Ali, J.S.; Qiblawey, Y.; Alassi, A.; Massoud, A.M.; Muyeen, S.M.; Abu-Rub, H. Power System Stability With High Penetration of Renewable Energy Sources: Challenges, Assessment, and Mitigation Strategies. *IEEE Access* **2025**, *13*, 39912–39934. [CrossRef]
8. Wang, Z.; Wei, W.; Pang, J.Z.F.; Liu, F.; Yang, B.; Guan, X.; Mei, S. Online Optimization in Power Systems With High Penetration of Renewable Generation: Advances and Prospects. *IEEE/CAA J. Autom. Sin.* **2023**, *10*, 839–858. [CrossRef]
9. Li, M.; Wang, Z.; Jia, M.; Liu, F.; Yang, B.; Guan, X. Distributed control to steer dynamical systems to the generalized Nash equilibria for monotone aggregative games with operational constraints. *Automatica* **2026**, *185*, 112794. [CrossRef]
10. Tinney, W.F.; Hart, C.E. Power Flow Solution by Newton's Method. *IEEE Trans. Power Appar. Syst.* **1967**, *PAS-86*, 1449–1460. [CrossRef]
11. Stott, B.; Alsac, O. Fast Decoupled Load Flow. *IEEE Trans. Power Appar. Syst.* **1974**, *PAS-93*, 859–869. [CrossRef]
12. de Moura, A.P.; de Moura, A.A.F. Newton–Raphson power flow with constant matrices: A comparison with decoupled power flow methods. *Int. J. Electr. Power Energy Syst.* **2013**, *46*, 108–114. [CrossRef]
13. Costilla-Enriquez, N.; Weng, Y.; Zhang, B. Combining Newton–Raphson and Stochastic Gradient Descent for Power Flow Analysis. *IEEE Trans. Power Syst.* **2021**, *36*, 514–517.
14. Borzacchiello, D.; Chinesta, F.; Malik, M.; García-Blanco, R.; Diez, P. Unified formulation of a family of iterative solvers for power systems analysis. *Electr. Power Syst. Res.* **2016**, *140*, 201–208. [CrossRef]
15. Baluev, D.; Ali, M.; Gryazina, E. State of the art approach for comprehensive power system security assessment—Real case study. *Int. J. Electr. Power Energy Syst.* **2024**, *155*, 109594. [CrossRef]
16. Han, Y.; Lu, C.; Song, W.; Wu, P.; Fang, H.; Liu, W. Review and Prospects of Online Calculation and Analysis Techniques for Power System Security and Stability. *Proc. CSEE* **2024**, *44*, 6733–6760. [CrossRef]
17. Bennani, H.; Chebak, A.; El Ouafi, A. A generic power flow formulation for flexible modeling and fast solving for large-scale unbalanced networks. *Int. J. Electr. Power Energy Syst.* **2023**, *148*, 108956. [CrossRef]

18. Balu, N.; Bertram, T.; Bose, A.; Brandwajn, V.; Cauley, G.; Curtice, D.; Fouad, A.; Fink, L.; Lauby, M.; Wollenberg, B.; et al. On-line power system security analysis. *Proc. IEEE* **1992**, *80*, 262–282.
19. Gholami, M.; Sanjari, M.J.; Safari, M.; Akbari, M.; Kamali, M.R. Static security assessment of power systems: A review. *Int. Trans. Electr. Energy Syst.* **2020**, *30*, e12432. [CrossRef]
20. Jadhav, S.; Sevak, B.; Das, S.; Su, W.; Bui, V.H. Enhancing Power Flow Estimation with Topology-Aware Gated Graph Neural Networks. *arXiv* **2025**, arXiv:2507.02078.
21. Chowdhury, B.H.; Wilamowski, B.M. Real-Time Power System Analysis Using Neural Computing. In *Proceedings of the Proceedings-Spie the International Society for Optical Engineering*; SPIE International Society For Optical: Bellingham, WA, USA, 1993; pp. 290–295.
22. Nguyen, T. Neural network load-flow. *IEE Proc. Gener. Transm. Distrib.* **1995**, *142*, 51–58. [CrossRef]
23. Donon, B.; Clément, R.; Donnot, B.; Marot, A.; Guyon, I.; Schoenauer, M. Neural networks for power flow: Graph neural solver. *Electr. Power Syst. Res.* **2020**, *189*, 106547. [CrossRef]
24. Liao, W.; Bak-Jensen, B.; Pillai, J.R.; Wang, Y.; Wang, Y. A Review of Graph Neural Networks and Their Applications in Power Systems. *J. Mod. Power Syst. Clean Energy* **2022**, *10*, 345–360. [CrossRef]
25. Lin, N.; Orfanoudakis, S.; Cardenas, N.O.; Giraldo, J.S.; Vergara, P.P. PowerFlowNet: Power flow approximation using message passing Graph Neural Networks. *Int. J. Electr. Power Energy Syst.* **2024**, *160*, 110112.
26. Wang, D.; Zheng, K.; Chen, Q.; Luo, G.; Zhang, X. Probabilistic Power Flow Solution with Graph Convolutional Network. In *Proceedings of the 2020 IEEE PES Innovative Smart Grid Technologies Europe (ISGT-Europe)*; IEEE: Piscataway, NJ, USA, 2020; pp. 650–654.
27. Shen, T.; Li, Y.; Xiang, J. A graph-based power flow method for balanced distribution systems. *Energies* **2018**, *11*, 511. [CrossRef]
28. Böttcher, L.; Wolf, H.; Jung, B.; Lutat, P.; Trageser, M.; Pohl, O.; Tao, X.; Ulbig, A.; Grohe, M. Solving AC Power Flow with Graph Neural Networks under Realistic Constraints. In *Proceedings of the 2023 IEEE Belgrade PowerTech*; IEEE: Piscataway, NJ, USA, 2023; pp. 1–7.
29. Veličković, P.; Cucurull, G.; Casanova, A.; Romero, A.; Lio, P.; Bengio, Y. Graph attention networks. *arXiv* **2017**, arXiv:1710.10903.
30. Li, H.; Liu, L.; Yu, S.; He, Q.; Wu, Q.; Zhang, J.; Lu, Q. Graph attention convolution network for power flow calculation considering grid uncertainty. *Int. J. Electr. Power Energy Syst.* **2025**, *165*, 110513. [CrossRef]
31. Kumar Mahto, D.; Bukya, M.; Kumar, R.; Mathur, A.; Kumar Saini, V. GAT-ADNet: Leveraging Graph Attention Network for Optimal Power Flow in Active Distribution Network With High Renewables. *IEEE Access* **2024**, *12*, 185728–185739.
32. Ngo, Q.H.; Nguyen, B.L.; Vu, T.V.; Zhang, J.; Ngo, T. Physics-informed graphical neural network for power system state estimation. *Appl. Energy* **2024**, *358*, 122602. [CrossRef]
33. Raissi, M.; Perdikaris, P.; Karniadakis, G.E. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *J. Comput. Phys.* **2019**, *378*, 686–707. [CrossRef]
34. El Semaan, E.; Iovine, A.; Le, T.D.; Dessante, P.; Rifi, M.; Croteau, D. A Physics-Informed-Neural-Network-based State Estimator for Low Voltage Distribution Systems. *IEEE Trans. Smart Grid* **2025**, *17*, 2544–2554.
35. Nellikkath, R.; Murzakhanov, I.; Chatzivasileiadis, S.; Venzke, A.; Bakhshizadeh, M.K. Physics-informed neural networks for phase locked loop transient stability assessment. *Electr. Power Syst. Res.* **2024**, *236*, 110790. [CrossRef]
36. Bento, M.E.C. Load margin assessment of power systems using physics-informed neural network with optimized parameters. *Energies* **2024**, *17*, 1562. [CrossRef]
37. Zheng, X.; Lee, A.; Hsien Huang, S.; Xie, L. A Physics-Informed Graph Neural Network Framework for N-2 Contingency Screening: A Real-World Texas Grid Study. *IEEE Open Access J. Power Energy* **2025**, *12*, 784–792. [CrossRef]
38. Yang, G.; Xu, G.; Jiang, R.; Jiang, Y.; Chen, X.; Sun, L.; Li, Y.; Gao, Y. A Parameter-Free Fault Location Algorithm for Hybrid Transmission Lines Using Double-Ended Data Synchronization and Physics-Informed Neural Networks. *Energies* **2025**, *18*, 5710. [CrossRef]
39. Lakshminarayana, S.; Sthapit, S.; Maple, C. Application of physics-informed machine learning techniques for power grid parameter estimation. *Sustainability* **2022**, *14*, 2051. [CrossRef]
40. Vairo, T.; Guarino, S.; Reverberi, A.P.; Fabiano, B. Physics-Informed Neural Networks for Thermal Anomaly Prediction in Battery Energy Storage Systems. *Energies* **2026**, *19*, 2503. [CrossRef]
41. Hu, X.; Hu, H.; Verma, S.; Zhang, Z.L. Physics-Guided Deep Neural Networks for Power Flow Analysis. *IEEE Trans. Power Syst.* **2021**, *36*, 2082–2092.
42. Nellikkath, R.; Chatzivasileiadis, S. Physics-Informed Neural Networks for Minimising Worst-Case Violations in DC Optimal Power Flow. In *Proceedings of the 2021 IEEE International Conference on Communications, Control, and Computing Technologies for Smart Grids (SmartGridComm)*; IEEE: Piscataway, NJ, USA, 2021; pp. 419–424.
43. Nellikkath, R.; Chatzivasileiadis, S. Physics-informed neural networks for ac optimal power flow. *Electr. Power Syst. Res.* **2022**, *212*, 108412. [CrossRef]

44. Jeddi, A.B.; Shafieezadeh, A. A Physics-Informed Graph Attention-based Approach for Power Flow Analysis. In *Proceedings of the 2021 20th IEEE International Conference on Machine Learning and Applications (ICMLA)*; IEEE: Piscataway, NJ, USA, 2021; pp. 1634–1640.
45. Zhang, H.F.; Lu, X.L.; Ding, X.; Zhang, X.M. Physics-informed line graph neural network for power flow calculation. *Chaos Interdiscip. J. Nonlinear Sci.* **2024**, *34*, 113123. [CrossRef]